COMPACTIFICATION DESCRIPTION OF COMPACT SUPPORT SECTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Editorial note: two light steps are asserted rather than displayed: the reduction to support containment is justified by the explicit nature of the covariance remark map, and the final support statement is called immediate from the vanishing of the stalks of j_!F at geometric points of the complement; both are definitional given the displayed data.

Allowed standard prerequisites (editorial).

- Compact support for proper schemes (Lemma lemma-proper-compact-support, canonical tag 0F73). cited at statement lines 612-613. Gives the equality H^0_c(X-bar, j_!F) = H^0(X-bar, j_!F) for the proper compactification.
- Restriction of j_! sheaves (Etale Cohomology, Lemma etale-cohomology-lemma-jshriek-open, canonical tag 0F70). cited at proof lines 617-618. (j_!F)|_X = F, identifying the displayed arrow as the covariance map.
- Covariance of compact support sections (Remark remark-covariance-compact-support, canonical tag 0F74). cited at proof lines 619-622. Supplies the injective map H^0_c(X, G|_X) -> H^0_c(X-bar, G) whose explicit form drives the reduction.

Source reference. Original lemma 0F73: sources/stacks-chow/tags.txt, source lines 18993–18993. Cited at statement lines 612-613: gives the equality H^0_c(\overline{X}, j_!F) = H^0(\overline{X}, j_!F) for the proper compactification.

Source reference. Original lemma 0F70: sources/stacks-chow/tags.txt, source lines 18990–18990. Cited at proof lines 617-618: (j_!F)|_X = F, identifying the displayed arrow as the covariance map.

Source reference. Original remark 0F74: sources/stacks-chow/tags.txt, source lines 18994–18994. Cited at proof lines 619-622: supplies the injective map H^0_c(X, G|_X) -> H^0_c(\overline{X}, G) whose explicit form drives the reduction.

Lemma 0.2. *Let k be a field. Let $j : X \to \overline{X}$ be an open immersion of schemes over k with $\overline{X}$ proper over k. For $\mathcal{F} \in Ab(X_{étale})$ there is a canonical isomorphism (see proof)*

$$H^0_c(X, \mathcal{F}) \longrightarrow H^0_c(\overline{X}, j_!\mathcal{F}) = H^0(\overline{X}, j_!\mathcal{F})$$

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where we have the equality on the right by Lemma 0F73.

Proof. We have $(j_!\mathcal{F})|_X = \mathcal{F}$, see Étale Cohomology, Lemma 0F70. Thus the displayed arrow is the injective map $H^0_c(X, \mathcal{G}|_X) \to H^0_c(\overline{X}, \mathcal{G})$ of Remark 0F74 for $\mathcal{G} = j_!\mathcal{F}$. The explicit nature of this map implies that it now suffices to show: if $s \in H^0(\overline{X}, \mathcal{G})$ is a section, then the support of s is contained in the open X. This is immediate from the fact that the stalks of $\mathcal{G}$ are zero at geometric points of $\overline{X} \setminus X$. $\square$